PRACTICE SHEET

1. Let A = {–1, 2, 5, 8}, B = {0, 1, 3, 6, 7} and R be the relation 'is one less than' from A to B, then how many elements will R contain?
(a) 2 (b) 3
(c) 5 (d) 9

2. If n(A) = 115, n(B) = 326, n(A – B) = 47, then what is n(A ∪ B) equal to?
(a) 373 (b) 165
(c) 370 (d) 394

3. If P(A) denotes the power set of A and A is the void set, then what is number of elements in P{P{P{P(A)}}}?
(a) 0 (b) 1
(c) 4 (d) 16

4. If $N_a = \{ax \mid x \in N\}$, then what is $N_{12} \cap N_8$ equal to?
(a) N_{12} (b) N_{20}
(c) N_{24} (d) N_{48}

5. If $X = \{(4^n - 3n - 1) \mid n \in N\}$ and $Y = \{9(n - 1) \mid n \in N\}$, then $X \cup Y$ equals to
(a) X (b) Y
(c) N (d) A null set

6. Sets A and B have n element in common. How many elements will (A × B) and (B × A) have in common
(a) 0 (b) 1
(c) n (d) n^2

7. What is the number of proper subsets of a given finite set with n elements?
(a) 2n – 1 (b) 2n – 2
(c) $2^n - 1$ (d) $2^n - 2$

8. If A = {a, b, c} and R = {(a, a), (a, b), (b, b), (c, c), (c, a)} is a binary relation on A, then which one of the following is correct?
(a) R is reflexive and symmetric but not transitive
(b) R is reflexive and transitive but not symmetric
(c) R is reflexive but neither symmetric nor transitive
(d) R is reflexive symmetric and transitive

9. If A, B and C are three finite sets, then $[(A \cup B) \cap C]'$ equals to
(a) $A' \cup B' \cap C'$ (b) $A' \cap B' \cap C'$
(c) $A' \cap B' \cup C'$ (d) $A \cap B \cap C$

10. If A and B are subsets of a set X, then what is the value of $\{A \cap (X - B)\} \cup B$?
(a) $A \cup B$ (b) $A \cap B$
(c) A (d) B

11. A = {1, 2, 3, 4} and R = {(1, 1), (1, 3), (2, 2), (3, 1), (3, 4), (4, 3), (4, 4)} is a relation on A × A, then which one of the following is correct?
(a) R is reflexive
(b) R is symmetric and transitive
(c) R is transitive but not reflexive
(d) R is neither reflexive nor transitive

12. Let R be the relation defined on the set of natural number N as aRb; b∈N, if a divides b. Then, which one of the following is correct?
(a) R is reflexive only (b) R is symmetric only
(c) R is transitive only (d) R is reflexive and transitive

13. Consider the following statements
I. $\phi \in \{\phi\}$ (II) $\{\phi\} \subseteq \phi$
Which of the statements given above is/are Correct?
(a) I only (b) II only
(c) Both I and II (d) Neither I nor II

14. Which one of the following is correct?
(a) $A \cup P(A) = P(A)$ (b) $A \cap P(A) = A$
(c) $A - P(A) = A$ (d) $P(A) - \{A\} = P(A)$
Here, P(A) denotes the power set of a set A.

15. If A, B and C are three sets, such that $A \cup B = A \cup C$, then which one of the following is correct?
(a) A = B only (b) B = c only
(c) A = C only (d) A = B + C

16. If A, B and C are three sets, then A – (B – C) equal
(a) $A - (B \cap C)$ (b) $(A - B) \cup C$
(c) $(A - B) \cup (A \cap C)$ (d) $(A - B) \cup (A - C)$

17. If a set A contains 4 elements, then what is the number of elements in A × P (A)?
(a) 16 (b) 32
(c) 64 (d) 128

18. If A and B are two sets satisfying A – B = B – A, then which one of the following is correct?
(a) $A = \phi$ (b) $A \cap B = \phi$
(c) A = B (d) None of these

19. For non – empty subsets A, B and C of a set X such that $A \cup B = B \cap C$, which one of the following is the strongest inference that can be derived?
(a) A = B = C (b) $A \subseteq B = C$
(c) $A = B \subseteq C$ (d) $A \subseteq B \subseteq C$

20. Let N be the set of integers. A relation R on N is defined as $R = \{(x, y) \mid xy > 0, y \in N\}$. Then, which one of the following is correct?
(a) R is symmetric but not reflexive
(b) R is reflexive but not symmetric
(c) R is symmetric and reflexive but not transitive
(d) R is an equivalence relation

21. If A and B are two disjoint sets, then which one of the following is correct?
(a) $A - B = A - (A \cap B)$
(b) $B - A' = B \cap A$

(c) $A \cap B = (A - B) \cap B$
(d) All of these

22. If the cardinality of a set A is 4 and that of a set B is 3, then what is the cardinality of the set $A \Delta B$?
(a) 1
(b) 5
(c) 7
(d) Cannot be determined as the sets A and B are not given.

23. The order of a Set A is 3 and that of a set B is 2. What is the number of relations from A to B?
(a) 4 (b) 6
(c) 32 (d) 64

24. If A = P{1, 2}, where P denotes the power set, then which one of the following is correct?
(a) $\{1, 2\} \subset A$ (b) $1 \in A$
(c) $\phi \notin A$ (d) $\{1, 2\} \in A$

25. Let N denote the set of natural numbers and $A = \{n^2 : a \in N\}$ and $\{n^3 : n \in N\}$. Which one of the following is correct?
(a) $A \cup B = N$
(b) The complement of $(A \cup B)$ is an infinite set
(c) $A \cap B$ must be a finite set
(d) $A \cap B$ must be a proper subset of $\{m6 : m \in N\}$

26. The relation R = {(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)} on a set A = {1, 2, 3} is
(a) Reflexive, transitive but not symmetric
(b) Reflexive, symmetric but not transitive
(c) Symmetric, transitive nut not reflexive
(d) Reflexive but neither symmetric nor transitive

ANSWER KEY

1.	b	**2.**	a	**3.**	d	**4.**	c	**5.**	b	**6.**	d	**7.**	c	**8.**	c	**9.**	c	**10.**	a
11.	d	**12.**	d	**13.**	d	**14.**	c	**15.**	b	**16.**	c	**17.**	c	**18.**	c	**19.**	d	**20.**	d
21.	d	**22.**	d	**23.**	d	**24.**	d	**25.**	d	**26.**	a								

Solutions

Sol.1. (b)
Given, A = {–1, 2, 5, 8} and B = {0, 1, 3, 6, 7}
$\therefore$ R = {(–1, 0), (2, 3), (5, 6)}
($\therefore$ A is one less than from B)

Sol.2. (a)
Now, $n(A - B) = n(A) - n(A \cap B)$
$\Rightarrow 47 = 115 - n(A \cap B)$
$\Rightarrow n(A \cap B) = 68$
$\therefore n(A \cup B) = n(A) + n(B) - n(A \cap B)$
$= 115 + 326 - 68 = 373$

Sol.3. (d)
The number of elements in power set of A is 1.
$[\therefore P(A) = 2^0 = 1]$
$\therefore P\{P(A)\} = 2^1 = 2$
$\Rightarrow P\{P\{P(A)\}\} = 2^2 = 4$
$\Rightarrow P\{P\{P\{P(A)\}\}\} = 2^4 = 16$

Sol.4. (c)
Given, $N_a = \{ax \mid x \in N\}$
$\therefore N_{12} = \{12, 24, 36, 48, \ldots\ldots.\}$
and $N_8 = \{8, 16, 24, \ldots\ldots.\}$
$\therefore N_8 \cap N_{12} = \{24, 48, \ldots\ldots.\}$
$= N_{24}$

Sol.5. (b)
$X = \{(4^n - 3n - 1) \mid n \in N\}$
and $Y = \{9(n - 1) \mid n \in N\}$
$\Rightarrow X = \{0, 9, 54, \ldots\ldots.\}$
and $Y = \{0, 9, 18, 27, 36, 54, \ldots\ldots\}$
$\therefore X \cup Y = \{0, 9, 18, 27, 36, 54, \ldots..\} = Y$

Sol.6. (d)
The total number of elements common in $(A \times B)$ and $(B \times A)$ is n^2 (by property)

Sol.7. (c)
Total number of proper subsets of a finite set with n elements $= 2^n - 1$.

Sol.8. (c)
$\therefore$ (a, a), (b, b), (c, c) $\in$R
$\therefore$ R is a reflexive relation.
But (a, b) $\in$R and (b, a) $\notin$ R,
$\therefore$ R is not a symmetric relation.
Also, (a, b), (b, c) $\in$ R
$\Rightarrow$ (a, c) $\notin$ R
$\therefore$ R is not a transitive relation.

Sol.9. (c)
We know that,
$[(A \cup B) \cap C'] = (A \cup B)' \cup C' = (A' \cap B') \cup C'$
$= A' \cap B' \cup C'$ (by De-Morgan's law)

Sol.10. (a)
$\therefore A \subseteq X$ and $B \subseteq X$
$\therefore \{(A \cap (X - B))\} \cup B$ $\quad (\therefore X - B = B')$
$= (A \cap B') \cup B = (A \cup B) \cap (B' \cup B)$
$= (A \cup B) \cap X = A \cup B$

Sol.11. (d)
Since, $3 \in A$
But $(3, 3) \notin A$
So, it is not reflexive.
and (3, 4) $\in$R and (4, 3) $\in$ R
But (3, 3) $\notin$R
So, it is also not transitive
Hence, R is neither reflexive nor transitive.

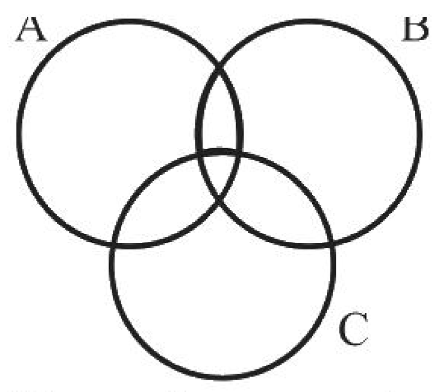
Sol.12. (d)
For reflexive
aRa $\Rightarrow$ a divides a
$\therefore$ R is reflexive.
For symmetric
aRb $\Rightarrow$ a divides b
bRa $\Rightarrow$ b divides a
which may not be possible.
$\therefore$ R is not symmetric.
For transitive
aRb $\Rightarrow$ a divides b $\Rightarrow$ b = ka
bRc $\Rightarrow$ b divides c $\Rightarrow$ $c = \ell b$
Now, $c = \ell ka$
$\Rightarrow$ a divides c $\Rightarrow$ aRc $\Rightarrow$ aRb, bRc $\Rightarrow$ cRa
Thus, R is transitive.

Sol.13. (a)
Only statement 1 is correct.

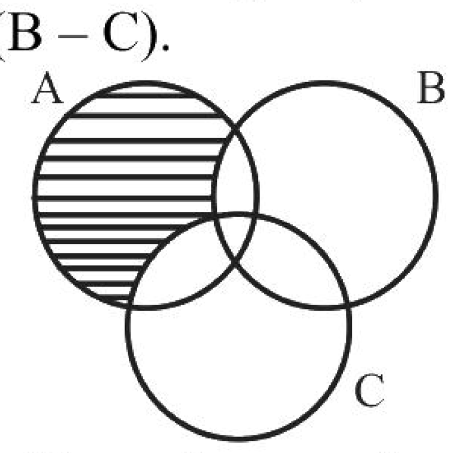
Sol.14. (c)
A – P (A) = A
Which is correct as A and P(A) are disjoint sets.

Sol.15. (b)
$\therefore$ Given that, $A \cup B = A \cup C$
and $A \cap B = A \cup C$
Then, let A = {a, b}, B = {a, c}
and C = {a, c}
$\Rightarrow A \cup B = \{a, b, c\}, A \cup C = \{a, b, c\}$
$A \cap B = \{a\}, A \cap C = \{a\}$,
$\therefore$ For B = C
$\Rightarrow A \cup B = A \cup C$
and $A \cap B = A \cap C$

Sol.16. (c)



In the above Venn diagram, shaded area shows $A - (B - C)$.



In the above Venn diagram, horizontal lines mean
$(A - B)$ and vertical lines mean $(A \cap C)$.
Total shaded portion = $(A - B) \cup (A \cap C)$
$\therefore (A - B) \cup (A \cap C) = A - (B - C)$

Sol.17. (c)
Since, the number of elements in set A is 4.
$\therefore$ Number of elements in $P(A) = 2^4 = 16$.
So, the number of elements in $A \times P(A) = 4 \times 16 = 64$.

Sol.18. (c)
$\therefore$ A and B are two sets satisfying $A - B = B - A$, which is possible only, if $A = B$.

Sol.19. (d)
$\therefore A \cup B = B \cap C$
From above strong inference is $A \subseteq B \subseteq C$.
e.g., $A = \{a\}$, $B = \{a, b\}$, $C = \{a, b, c\}$

Sol.20. (d)
$\therefore R = \{(x, y) \mid xy > 0, x, y \in N\}$
Reflexive
$\therefore x, y \in N$
$\therefore x, x \in N \Rightarrow x^2 > 0$
Symmetric
$\therefore x, y \in N$
and $xy > 0 \Rightarrow yx > 0$
$\therefore$ R is also symmetric.
Transitive
$\therefore x, y, z \in N \Rightarrow xy > 0, yz > 0 \Rightarrow xz > 0$
$\therefore$ R is also transitive.

Sol.21. (d)
$\therefore A \cap B = \phi$
$\therefore A - B = A - (A \cap B) \quad (\therefore A - B = A)$
Now, $B - A' = \phi$
and $B \cap A = \phi$
or $B - A' = B \cap A$
and $(A - B) \cap B = A \cap B$

Sol.22. (d)
Since, the sets A and B are not known, then cardinality of the set $A \Delta B$ cannot be determined.

Sol.23. (d)
$\therefore n(A) = 3$ and $n(B) = 2$
$\therefore$ Number of relations from A to B = $2^{[n(A) \times n(B)]}$
$= 2^{(3 \times 2)} = 2^6 = 64$

Sol.24. (d)
$A = P(\{1, 2\}) = \{\phi, \{1\}, \{2\}, \{1, 2\}\}$
From above, it is clear that
$\{1, 2\} \in A$

Sol.25. (d)
$\therefore A = \{n^2 : n \in N\}$ and $B = \{n^3 : n \in N\}$
$A = \{1, 4, 9, 16, 25, 49, 64, 81, \ldots\ldots\}$
$B = \{1, 8, 27, 64, 125, \ldots\ldots\}$
$A \cap B = \{1, 64, \ldots.\}$
$\therefore A \cap B$ must be a proper subset of $\{m^6 : m \in N\}$

Sol.26. (a)
$\therefore R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$
Reflexive
$\therefore$ 1R1, 2R2, 3R3
$\therefore$ R is a reflexive relation.
Symmetric
$\therefore$ 1R2 but 2R1
$\therefore$ R is not a symmetric relation.
Transitive
$\therefore$ 1R2, 2R3 $\Rightarrow$ 1R3
$\therefore$ R is a transitive relation.
thus, R is an equivalence relation.

NDA PYQ

1. If A = {1, 2, 5, 6} and B = {1, 2, 3}, then what is (A × B) ∩ (B × A) equal to?
(a) {(1, 1), (2, 1), (6, 1), (3, 2)}
(b) {(1, 1), (1, 2), (2, 1), (2, 2)}
(c) {(1, 1), (2, 2)}
(d) {(1, 1), (1, 2), (2, 5), (2, 6)}
[NDA – 2011(1)]

2. If A, B and C are non-empty sets such that A ∩ C = ϕ, then what is (A × B) ∩ (C × B) equal to?
(a) A × C (b) A × B
(c) B × C (d) ϕ
[NDA - 2011(1)]

3. If A = {4n + 2| n is natural number} and B = {3n | n is a natural number}, then what is (A ∩ B) equal to?
(a) $\{12n^2 + 6n \mid n \text{ is a natural number}\}$
(b) {24 n – 12 | n is a natural number}
(c) $\{12n^2 - 6n \mid n \text{ is a natural number}\}$
(d) {12n – 6 | n is a natural number}
[NDA - 2011(1)]

4. Consider the following with regard to a relation R on a set of real numbers defined by xRy if and only if 3x + 4y = 5

I. 0R1 II. $1R\frac{1}{2}$ III. $\frac{2}{3}R\frac{3}{4}$
Which of the above are correct?
(a) I and II (b) I and III
(c) II and III (d) I, II and III
[NDA - 2011(1)]

5. Let M be the set of men and R is a relation 'is son of defined on M. Then, R is
(a) An equivalence relation
(b) A symmetric relation only
(c) A transitive relation only
(d) None of the above
[NDA - 2011(1)]

6. If P, Q, R are three non-collinear points, then what is PQ ∩ PR equal to?
(a) Null set (b) {P}
(c) {P, Q, R} (d) {Q, R}
[NDA-2011(1)]

7. What is the set of points (x, y) satisfying the equations $x^2 + y^2 = 4$ and x + y = 2?
(a) {(2,0), (–2,0), (0,2)}
(b) {(0,2), (0,–2)}
(c) {(0,2), (2,0)}
(d) {(2,0), (–2,0), (0,2), (0,–2)}
[NDA-2011(1)]

Direction (for next five)
The Students of a class are offered three language (Hindi, English and French). 15 students learn all the three language whereas 28 students do not learn any language. The number of students learning Hindi and English but not French is twice the number of students learning Hindi and French but not English. The number of students learning English and French but not Hindi is thrice the number of students learning Hindi and French but not English. 23 students learn only Hindi and 17 students learn only English. The total number of students learning French is 46 and the total number of students learning only French is 11.

8. How many students learn precisely two languages?
(a) 55 (b) 40
(c) 30 (d) 13
[NDA - 2011(1)]

9. How many students learn at least two languages?
(a) 15 (b) 30
(c) 45 (d) 55
[NDA - 2011(1)]

10. What is the total strength of the class?
(a) 124 (b) 100
(c) 96 (d) 66
[NDA - 2011(1)]

11. How many students learn English and French?
(a) 30 (b) 43
(c) 45 (d) 73
[NDA - 2011(1)]

12. How many students learn at least one language?
(a) 45 (b) 51
(c) 96 (d) None of these
[NDA - 2011(1)]

13. If ϕ is a null set, then which one of the following is correct?
(a) ϕ = 0 (b) ϕ = {0}
(c) ϕ = {ϕ} (d) ϕ = { }
[NDA - 2011(2)]

14. If A = {a, b, c}, then what is the number of proper subsets of A?
(a) 5 (b) 6
(c) 7 (d) 8
[NDA - 2011(2)]

15. If A and B are any two sets, then what is the value of A ∩ (A ∪ B)?
(a) Complement of A (b) Complement of B
(c) B (d) A
[NDA-2012(1)]

16. Let U = {x ∈ N: 1 ≤ x ≤ 10} be the universal set, N being the set of natural numbers If A = {1, 2, 3, 4} and B = {2, 3, 6, 10}, then what is the complement of (A – B)?
(a) {6, 10} (b) {1, 4}
(c) {2, 3, 5, 6, 7, 8, 9, 10} (d) {5, 6, 7, 8, 9, 10}
[NDA-2012(1)]

17. Let A = {x : x is a square of a natural number and x is less than 100} and B is set of even natural numbers. What is the cardinality of A ∩ B?
(a) 4 (b) 5
(c) 9 (d) None of these
[NDA-2012(1)]

18. Out of 500 first year students, 260 passed in the first semester and 210 passed in the second semester. If 170 did not pass in either semester, how many passed in both semesters?
(a) 30 (b) 40
(c) 70 (d) 140
[NDA-2012(1)]

19. If A and B are two non-empty sets having n elements in common, then what is the number of common elements in the sets A×B and B×A?
(a) n (b) n^2
(c) 2n (d) zero
[NDA-2012(1)]

20. If $A=\{0,1\}$ and $B=\{1,0\}$, then what is $A\times B$ equal to?
(a) $\{(0,1), (1,0)\}$ (b) $\{(0,0), (1,1)\}$
(c) $\{(0,1), (1,0), (1,1)\}$ (d) $A \times A$
[NDA-2012(1)]

For the next Four (04) question that follow:
In a city, three daily newspapers A, B, C are published, 42% read A; 51% read B; 68% read C; 30% read A and B; 28% read B and C; 36% read A and C; 8% do not read any of the three newspapers.

21. What is the percentage of persons who read all the three papers?
(a) 20% (b) 25%
(c) 30% (d) 40%
[NDA-2012(1)]

22. What is the percentage of persons who read only two papers?
(a) 19% (b) 31%
(c) 44% (d) none of the above
[NDA-2012(1)]

23. What is the percentage of persons who read only one paper?
(a) 38% (b) 48%
(c) 51% (d) none of the above
[NDA-2012(1)]

24. What is the percentage of person who read only A but neither B nor C?
(a) 4% (b) 3%
(c) 1% (d) none of the above
[NDA-2012(1)]

25. The set $A=\{x : x + 4 = 4\}$ can also be represented by:
(a) 0 (b) φ
(c) (φ) (d) $\{0\}$
[NDA-2012(1)]

26. The relation 'has the same father as' over the set of children is:
(a) only reflexive (b) only symmetric
(c) only transitive (d) an equivalence relation
[NDA-2012(2)]

27. Let $P=\{1,2,3\}$ and a relation on set P is given by the set $R=\{(1,2), (1,3), (2,1), (1,1), (2,2), (3,3), (2,3)\}$. Then R is:
(a) Reflexive, transitive but not symmetric
(b) Symmetric, transitive but not reflexive
(c) Symmetric, reflexive but not transitive
(d) None of the above
[NDA-2012(2)]

28. If non-empty set A contains n elements, then its power set contains how many elements?
(a) n^2 (b) 2^n
(c) $2n$ (d) $n+1$
[NDA-2012(2)]

29. Let $A=\{x\in W$, the set of whole numbers and $x<3\}$, $B=\{x\in N$, the set of natural numbers and $2 \le x < 4\}$ and $C=\{3,4\}$, then how many elements will $(A\cup B) \times C$ contain?
(a) 6 (b) 8
(c) 10 (d) 12
[NDA-2012(2)]

30. Which one of the following is a null set?
(a) $\{0\}$
(b) $\{ \{ \{\} \} \}$
(c) $\{ \{\} \}$
(d) $\{x \mid x^2 + 1 = 0, x\in R\}$
[NDA – 2013(1)]

Direction: For the next six (06) questions that follows:
In a state with a population of 75×10^6, 45% of them know Hindi, 22% know English, 18% know Sanskrit, 12% know Hindi and English, 8% know English and Sanskrit, 10% know Hindi and Sanskrit and 5% know all the three languages:

31. What is the number of people who do not know any of the above three languages?
(a) 3×10^6 (b) 4×10^6
(c) 3×10^7 (d) 4×10^7
[NDA-2013(1)]

32. What is the number of people who know Hindi only?
(a) 21×10^6 (b) 25×10^6
(c) 28×10^6 (d) 3×10^7
[NDA-2013(1)]

33. What is the number of people who know Sanskrit only?
(a) 5×10^5 (b) 45×10^6
(c) 4×10^6 (d) none of the above
[NDA-2013(1)]

34. What is the number of people who know English only?
(a) 5×10^6 (b) 45×10^6
(c) 4×10^6 (d) none of these
[NDA-2013(1)]

35. What is the number of people who know only one language?
(a) 3×10^6 (b) 4×10^6
(c) 3×10^7 (d) 4×10^7
[NDA-2013(1)]

36. What is the number of people who know only two languages?
(a) 11.25×10^5 (b) 11.25×10^6
(c) 12×10^5 (d) 12.5×10^5
[NDA-2013(1)]

37. If $A = \{x,y\}$, $B = \{2,3\}$, $C = \{3,4\}$, then what is the number of elements in $A\times(B\cup C)$?
(a) 2 (b) 4
(c) 6 (d) 8
[NDA-2013(1)]

38. If A is a relation on a set R, then which one of the following is correct?
(a) $R\subseteq A$ (b) $A\subseteq R$
(c) $A\subseteq(R\times R)$ (d) $R\subseteq(A\times A)$
[NDA-2013(1)]

39. If A is a subset of B, then which one of the following is correct?
(a) $A^C \subseteq B^C$ (b) $B^C \subseteq A^C$
(c) $A^C = B^C$ (d) $A \subseteq A \cap B$
[NDA – 2013(2)]

40. If $A = \{1, 3, 5, 7\}$, then what is the cardinality of the power set P(A)?
(a) 8 (b) 15
(c) 16 (d) 17
[NDA – 2013(2)]

41. Consider the following
I. $A \cup (B \cap C) = (A \cap B) \cup (A \cap C)$
II. $A \cap (B \cup C) = (A \cup B) \cap (A \cup C)$
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) I and II (d) Neither I nor II
[NDA – 2013(2)]

42. In a group of 50 people, two tests were conducted, one for diabetes and one for blood pressure. 30 people were diagnosed with diabetes and 40 people were diagnosed with high blood pressure. What is the minimum number of people who were having diabetes and high blood pressure?
(a) 0 (b) 10
(c) 20 (d) 30

[NDA – 2013(2)]

43. Which one of the following is an example of non-empty set?

(a) Set of all even prime numbers
(b) $\{x : x^2 - 2 = 0$ and x is rational$\}$
(c) $\{x : x$ is a natural number, $x < 8$ and simultaneously $x > 12\}$
(d) $\{x : x$ is a point common to any two parallel lines$\}$

[NDA – 2013(2)]

44. If A={1,2}, B=(2,3) and C={3,4}, then what is the cardinality of $(A\times B)\cap(A\times C)$?
(a) 8 (b) 6
(c) 2 (d) 1

[NDA-2013(2)]

45. If A is a finite set having n elements, then the number of relations which can be defined in A is:
(a) 2^n (b) n^2
(c) 2^{n^2} (d) n^n

[NDA-2013(2)]

46. The relation R in the set Z of integers given by R={(a, b): a–b is divisible by 5} is:
(a) reflexive
(b) reflexive but not symmetric
(c) symmetric and transitive
(d) an equivalence relation

[NDA-2013(2)]

47. Let A= {a, b, c, d} and B = {x, y, z}. What is the number of elements in $A\times B$?
(a) 6 (b) 7
(c) 12 (d) 64

[NDA-2013(2)]

Direction (for next three): Read the following information carefully to answer the questions that follow.
In a survey of 25 students, it was found that 15 have taken Mathematics, 12 have taken Physics and 11 taken Chemistry, 5 have taken Mathematics and Chemistry, 9 have taken Mathematics and Physics, 4 have taken Physics and Chemistry and 3 have taken all the three subjects.

48. The number of students who have taken only Physics, is

(a) 2 (b) 3
(c) 5 (d) 6

[NDA – 2014(1)]

49. The number of students who have taken only two subjects, is

(a) 7 (b) 8
(c) 9 (d) 10

[NDA – 2014(1)]

50. Consider the following statements
I.The number of students who have taken only one subject is equal to the number of students who have taken only two subjects.
II.The number of students who have taken at least two subjects is four times the number of students who have taken all the three subjects.
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA – 2014(1)]

51. Let X be the set of all citizens of India. Elements x, y in X are said to be related if the difference of their age is 5 years. Which one of the following is correct?
(a)the relation is an equivalence relation of X.
(b)the relation is symmetric but neither reflexive nor transitive
(c)the relation is reflexive but neither symmetric nor transitive
(d)none of the above

[NDA-2014(1)]

52. Let S denote set of all integers. Define a relation R on S as ‘aRb if $ab \geq 0$ where $a, b\in S$’. Then R is:
(a)Reflexive but neither symmetric nor transitive relation
(b)Reflexive, symmetric but not transitive relation
(c)An equivalence relation
(d)Symmetric but neither reflexive nor transitive relation

[NDA-2014(1)]

53. The relation S is defined on the set of integers Z as xSy if integer x divides integer y. Then:
(a) S is an equivalence relation
(b) S is only reflexive and symmetric
(c) S is only reflexive and transitive
(d) S is only symmetric and transitive

[NDA-2014(2)]

54. A and B are two sets having 3 elements in common. If n(A)=5, n(B)=4, then what is $n(A\times B)$ equal to?
(a) 0 (b) 9
(c) 15 (d) 20

[NDA-2014(2)]

55. In a class of 60 students, 45 students like music, 50 students like dancing, 5 students like neither. Then, the number of students in the class who like both music and dancing, is
(a) 35 (b) 40
(c) 50 (d) 55

[NDA – 2015(1)]

56. Let A = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. Then, the number of subsets of A containing exactly two elements is
(a) 20 (b) 40
(c) 45 (d) 90

[NDA – 2015(1)]

57. If A = {x : x is a multiple of 3} and B = {x : x is a multiple of 4} and C = {x : x is a multiple of 12}, then which one of the following is null set?
(a) $(A \setminus B) \cup C$ (b) $(A \setminus B) \setminus C$
(c) $(A \cap B) \cap C$ (d) $(A \cap B) \setminus C$

[NDA 2015(1)]

58. Let X be the set of all persons living in a city. Persons x, y in X are said to be related as $x<y$ if y is at least 5 years older than x. Which one of the following is correct?
(a)the relation is an equivalence relation on X.
(b)the relation is transitive but neither reflexive nor symmetric
(c)the relation is reflexive but neither transitive nor symmetric
(d)the relation is symmetric but neither transitive nor reflexive

[NDA-2015(1)]

59. Let Z be the set of integers and aRb, where $a, b \in Z$ if and only if (a–b) is divisible by 5.
Consider the following statements:
1.The relation R partitions Z into five equivalent classes.
2.Any two equivalent classes are either equal or disjoint
Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA-2015(1)]

60. Let A={x, y, z} and B={p, q, r, s}. What is the number of distinct relations from B to A?

(a) 4096 (b) 4094
(c) 128 (d) 126

[NDA-2015(1)]

61. If A = {x ∈ R : $x^2 + 6x - 7 < 0$} and B = {x ∈ R : $x^2 + 9x + 14 > 0$}, then which of the following is/are correct?
I. (A ∩ B) = (–2, 1)
II. (A \ B) = (–7, –2)
Select the correct answer using the code given below.

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA – 2015(2)]

62. A, B, C and D are four sets such that A ∩ B = C ∩ D = ϕ. Consider the following
I. A ∪ C and B ∪ D are always disjoint.
II. A ∩ C and B ∩ D are always disjoint.
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA – 2015(2)]

63. Let X be the set of all persons living in Delhi. The person a and b in X are said to be related if the difference in their ages is at most 5 years. The relation is:
(a) an equivalence relation
(b) reflexive and transitive but not symmetric
(c) symmetric and transitive but not reflexive
(d) reflexive and symmetric but not transitive

[NDA-2015(2)]

64. Suppose there is a relation * between the positive numbers x and y given by x*y if and only if $x \le y^2$. Then which one of the following is correct?
(a) * is reflexive but not transitive and symmetric
(b) *is transitive but not reflexive and symmetric
(c) *is symmetric and reflexive but not transitive
(d) *is symmetric but not reflexive and transitive

[NDA-2016(1)]

65. Let R be a relation on the set N of natural numbers defined by 'nRm ⇔ n is a factor of m'. Then which one of the following is correct?
(a) R is reflexive, symmetric but not transitive
(b) R is transitive, symmetric but not reflexive
(c) R is reflexive, transitive but not symmetric
(d) R is an equivalence relation

[NDA-2016(1)]

66. What is the number of natural numbers less than or equal to 1000 which are neither divisible by 10 nor 15 nor 25?
(a) 860 (b) 854
(c) 840 (d) 824

[NDA-2016(1)]

67. Let S be a set of all distinct numbers of the form $\frac{p}{q}$, where p, q ∈ (1, 2, 3, 4, 5, 6). What is the cardinality of the ser S?
(a) 21 (b) 23
(c) 32 (d) 36

[NDA – 2016(2)]

68. If A = {x∈R : $x^2 + 6x - 7 < 0$} and B = {x∈R: $x^2 + 9x + 14 > 0$}, then which of the following is/are correct?
1. A ∩ B = {x∈R: –2 < x < 1}
2. A\B = {x∈R: –7 < x < –2}
Select the correct answer using the codes given below:
(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA – 2016(2)]

69. A coin is tossed three times. Consider the following events:
A: No head appears
B: Exactly one head appears
C: At least two heads appear
Which one of the following is correct?
(a) (A ∪ B) ∩ (A ∪ C) = B ∪ C
(b) (A ∩ B') ∪ (A ∩ C') = B' ∪ C'
(c) A ∩ (B' ∪ C') = A ∪ B ∪ C'
(d) A ∩ (B' ∪ C') = B' ∩ C'

[NDA – 2016(2)]

70. Let R be a relation from A={1,2,3,4} to B={1,3,5} such that: R=[(a,b): a<b, where a∈A and b∈B] what is RoR^{-1} equal to?
(a) {(1,3), (1,5), (2,3), (2,5), (3,5), (4,5)}
(b) {(3,1), (5,1), (3,2), (5,2), (5,3), (5,4)}
(c) {(3,3), (3,5), (5,3), (5,5)}
(d) {(3,3), (3,4), (4,5)}

[NDA-2016(2)]

71. In an examination, 70% students passed in Physics, 80% students passed in Chemistry, 75% students passed in Mathematics and 85% students passed in Biology, and x% students failed in all the four subject. What is the minimum value of x?
(a) 10 (b) 12
(c) 15 (d) none of the above

[NDA-2016(2)]

72. If S = {x : $x^2 + 1 = 0$, x is real}, then S is:
(a) {–1} (b) {0}
(c) {1} (d) An empty set

[NDA – 2017(1)]

73. Let S be the set of all persons living in Delhi. We say that x, y in S are related if they were born in Delhi on the same day. Which one of the following is correct?
(a) The relation is an equivalent relation
(b) The relation is not reflexive but it is symmetric and transitive
(c) The relation is not symmetric but it is reflexive and transitive
(d) The relation is not transitive but it is reflexive and symmetric

[NDA – 2017(1)]

74. Consider the following in respect of sets A and B:
1. (A – B) ∪ B = A
2. (A – B) ∪ A = A
3. (A – B) ∩ B = ϕ
4. A ⊆ B ⇒ A ∪ B = B
Which of the above are correct?
(a) 1, 2 and 3 (b) 2, 3 and 4
(c) 1, 3 and 4 (d) 1, 2 and 4

[NDA – 2017(1)]

75. If E is the universal set and A = B ∪ C, then the set E – (E – (E – (E – (E – A)))) is same as the set:
(a) B' ∪ C' (b) B ∪ C
(c) B' ∩ C' (d) B ∩ C

[NDA – 2017(2)]

76. If A = {x : x is a multiple of 2}, B = {x : x is a multiple of 5} and C = {x : x is a multiple of 10}, then A ∩ {B ∩ C} is equal to:
(a) A
(b) B
(c) C
(d) {x : x is a multiple of 100}

[NDA – 2017(2)]

77. If we define a relation R on the set N > as (a, b) R (c, d) ⇔ a + d = b + c for (a, b), (c, d) ∈ N × N, then the relation

(a) symmetric only
(b) symmetric and transitive only
(c) equivalence relation
(d) reflexive only

[NDA-2017(2)]

78. A survey of 850 students in a University yield that 680 students like music and 215 like dance. What is the least number of student who like both music and dance?
(a) 40 (b) 45
(c) 50 (d) 55

[NDA-2018(1)]

Direction (for next two):
In a class, 54 students are good in Hindi only, 63 students are good in Mathematics only and 41 students are good in English only. There are 18 students who are good in both Hindi and Mathematics. 10 students are good in all three subjects.

79. What is the number of students who are good in either Hindi or Mathematics but not in English?
(a) 99 (b) 107
(c) 125 (d) 130

[NDA (I) - 2018]

80. What is the number of students who are good in Hindi and Mathematics but not in English?
(a)18 (b) 12
(c) 10 (d) 8

[NDA (I) - 2018]

81. Let A and B be subsets of X and C = $(A\cap B')\cup(A'\cap B)$ where A' and B' are complements of A and B respectively in X. What is C equal to?
(a) $(A\cup B')-(A\cap B')$ (b) $(A'\cap B)-(A'\cap B)$
(c) $(A\cup B)-(A\cap B)$ (d) $(A'\cup B')-(A'\cap B')$

[NDA (I) - 2018]

Direction for (for next two):
A survey was conducted among 300 students. It was found that 125 students like to play cricket, 145 students like to play football and 90 students like to play tennis, 32 students like to play exactly two games out of the three games.

82. How many students like to play all three games?
(a) 14 (b) 21
(c) 28 (d) 35

[NDA (II) - 2018

83. How many students like to play exactly only one game?
(a)196 (b) 228
(c) 254 (d) 268

[NDA (II) - 2018

84. If A, B and C are subsets of a universal set, then which of the following is not correct?
(a) $A\cup(B\cap C) = (A\cup B) \cap (A\cup C)$
(b) $A'\cup(A\cup B) = (B'\cap A)' \cup A$
(c) $A'\cup(B\cup C) = (C'\cap B)' \cap A'$
(d) $(A\cap B)\cup C = (A\cup C)\cap(B\cup C)$
Where A' is complement of A

[NDA (II) - 2018]

Direction for (next two):
In a school, all the students play atleast one of the games - chess, carrom and table tennis. 60 play chess, 50 play table tennis, 48 play carrom, 12 play chess and carrom, 15 play carrom and table tennis, 20 play table tennis and chess.

85. What can be the minimum number of students in the school?
(a) 123 (b) 111
(c) 95 (d) 63

[NDA (I) - 2019]

86. What can be the maximum number of students in the school?
(a) 111 (b) 123
(c) 125 (d) 135

[NDA (I) - 2019]

87. Suppose X = {1,2,3,4} and R is a relation on X. If R = {(1,1),(2,2),(3,3),(1,2),(2,1),(2,3), (3,2)}, then which one of the following is correct ?
(a) R is reflexive and symmetric, but not transitive
(b) R is symmetric and transitive, but not reflexive
(c) R is reflexive and transitive, but not symmetric
(d) R is neither reflexive butnor transitive,symmetric

[NDA (I) - 2019]

88. A relation R is defined on the set N of natural numbers as $xRy \Rightarrow x^2 - 4xy + 3y^2 = 0$. Then which one of the following is correct?
(a) R is reflexive and symmetric, but not transitive
(b) R is symmetric and transitive, but not reflexive
(c) R is reflexive, symmetric and transitive
(d) R is reflexive, but neither symmetric nor transitive

[NDA (I) - 2019]

89. Consider the following statements for the non- empty sets A and B:
1. $(A \cap B) \cup (A \cap B') \cup (A' \cap B) = A \cup B$
2. $(A \cup (A' \cap B')) = A \cup B$
Which of the following statements is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2019]

90. Let X be a non-empty set and let A, be subsets of X. Consider the following statements:
1. $A \subset C \Rightarrow (A \cap B) \subset (B \cap C),(A \cup B) \subset (C \cup B)$
2. $(A \cap B) \subset (C \cap B)$ for all sets $B \Rightarrow A \subset C$
3. $(A \cup B) \subset (C \cup B)$ for all sets $B \Rightarrow A \subset C$
Which of the following statements is/are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA (I) - 2019]

91. If $A = \{\lambda,\{\lambda, \mu\}\}$, then the power set of A is:
(a) $\{\phi,\{\phi\}, \{\lambda\}, \{\lambda, \mu\}\}$
(b) $\{\phi,\{\lambda\}, \{\{\lambda, \mu\}\}, \{\lambda,\{\lambda, \mu\}\}\}$
(c) $\{\phi, \{\lambda\}, \{\lambda, \mu\}, \{\lambda, \{\lambda, \mu\}\}\}$
(d) $\{\{\lambda\}, \{\lambda, \mu\}, \{\lambda, \{\lambda, \mu\}\}\}$

[NDA-2019(1)]

92. Let S = {2,4,6,............20}, what is the number of subsets does S have ?
(a) 10 (b) 20
(c) 512 (d) 1024

[NDA (II) - 2019]

93. If A, B and C are subsets of a given set, then which one of the following is relations is not correct?
(a) $A\cup (A\cap B) = A\cup B$
(b) $A\cap(A\cup B) = A$
(c) $(A\cap B)\cup C = (A\cup C)\cap(B\cup C)$
(d) $(A\cup B)\cap C = (A\cap C)\cup(B\cap C)$

[NDA (II) - 2019]

94. If a set A contains 3 elements and another set B contains 6 elements, then what is the minimum number of elements that $(A\cup B)$ can have?
(a) 3 (b) 6
(c) 8 (d) 9

[NDA (II) - 2019]

95. In a school, 50% students play cricket and 40% play football. If 10% of students play both the games, then what percent of students play neither cricket nor football?

(a) 10% (b) 15%
(c) 20% (d) 25%

[NDA (II) - 2019]

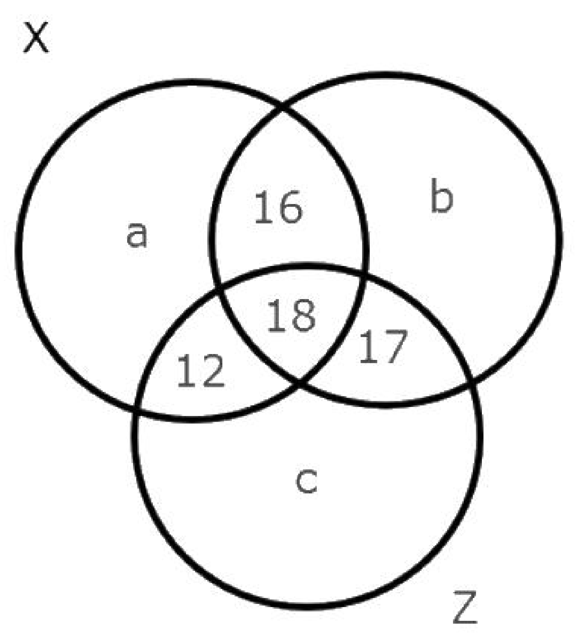
96. If A = {x: 0 ≤ x ≤ 2} and B = {y: y is a prime number}, then what is A∩B equal to ?
(a) ∅ (b) {1}
(c) {2} (d) {1,2}

[NDA (II) - 2019]

97. Consider the proper subsets of {1,2,3,4}. How many of these proper subsets are superset of the set {3}?
(a) 5 (b) 6
(c) 7 (d) 8

[NDA 2020]

Direction (For next three):
Consider the following Venn diagram, where X, Y and Z are three sets. Let the number of elements in Z is denoted by n(Z) which is equal to 90.



98. If the number of elements in Y and Z are in the ratio 4:5, then what is the value of b?
(a) 18 (b) 19
(c) 21 (d) 23

[NDA 2020]

99. What is the value of n(X) + n(Y)+n(Z) −n(X∩Y) −n(Y∩Z) −n(Z∩X)+ n(X∩Y∩Z)?
(a) a + b +43 (b) a + b + 63
(c) a + b +96 (d) a + b +106

[NDA 2020]

100. If the number of elements belonging to neither X, nor Y, nor Z is equal to p, then what is the number of elements in the complement of X?
(a) p + b + 60 (b) p + b + 40
(c) p + a + 60 (d) p + a + 40

[NDA 2020]

101. Let S = {1,2,3,....}. A relation R on S×S is defined by xRy if $\log_a x > \log_a y$, when a = 1/2 , The relation is
(a) reflexive only
(b) symmetric only
(c) transitive only
(d) both symmetric and transitive

[NDA 2020]

102. Consider the following statements:
1. A = {1,3,5} and B = {2,4,7} are equivalent sets.
2. A = {1,5,9} and B = {1,5,5,9,9} are equal sets.
Which of the above statement(s) is/are correct?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

[NDA (I) 2021]

103. Consider the following statements:
1. The null set is a subset of every set.
2. Every set is a subset of itself.
3. If a set has 10 elements, then its power set will have 1024 elements.
Which of the above statement(s) is/are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1,2 and 3

[NDA (I) 2021]

104. Let R be a relation defined as xRy if and only if 2x + 3y = 20, where x,y∈N. How many elements of the form (x,y) are there in R ?
(a) 2 (b) 3
(c) 4 (d) 6

[NDA (I) 2021]

105. Consider the following statements in respect of sets:
1.The union over intersection of sets is distributive
2.The complement of union of two sets is equal to intersection of their complements
3.If the difference of two sets is equal to empty set, then the two sets must equal.
(a)1 and 2 only (b) 2 and 3 only
(c)1 and 3 only (d) 1, 2 and 3

[NDA (II) 2021]

106. Consider three sets X, Y and Z having 6, 5 and 4 elements respectively. All these 15 elements are distinct. Let S = (X – Y) ∪ Z. How many proper subsets does S have?
(a) 255 (b) 256
(c) 1023 (d) 1024

[NDA (II) 2021]

107. Suppose set A consists of first 250 natural numbers that are multiples of 3 and set B consists of first 200 even natural numbers. How many elements does A ∪ B have?
(a) 324 (b) 364
(c) 384 (d) 400

[NDA (II) 2021]

108. Consider all the subsets of the set A = {1,2,3,4}. How many of them are superset of the set {4}?
(a) 6 (b) 7
(c) 8 (d) 9

[NDA (I) - 2022]

109. Consider the following statements in respect of two non-empty sets A and B?
1.x∉ (A ⋃ B) ⇒ x ∉ A or x ∉ B
2.x∉ (A ⋂ B) ⇒ x ∉ A and x ∉ B
Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2022]

110. Consider the following statements in respect of two non-empty sets A and B:
1. A ⋃ B = A ⋂ B if A = B
2. A Δ B = φ if A = B
Which of the above statement is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA (I) - 2022]

111. Consider the following statements in respect of the relation R in the set IN of natural numbers defined by xRy if $x^2 – 5xy + 4y^2 = 0$.
1.R is reflexive
2.R is symmetric
3.R is transitive
Which of the above statement is/are correct?
(a) 1 only (b) 2 only
(c) 1 and 2 only (d) 1, 2 and 3

[NDA (I) - 2022]

112. Consider the following statements in respect of any relation R on a set A:
1.If R is reflexive, then R^{-1} is also reflexive

2.If R is symmetric, then R^{-1} is also symmetric
3.If R is transitive, then R^{-1} is also transitive
Which of the above statements are correct?
(a)1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA (I) - 2022]

113. If A = {{1, 2, 3}}, then how many elements are there in the power set of A?
(a) 1 (b) 2
(c) 4 (d) 8

[NDA (I) - 2022]

Consider the following for the next three (03) items that follow:
A university awarded medals in basket ball, football and volleyball. Only x students (x < 6) got medal in all the three sports and the medals went to a total of 15x students. It awarded 5x medals in basketball, (4x + 15) medals in football and (x + 25) medals in volleyball.

114. How many received medals in exactly two of the three sports?
(a) 30 – 4x (b) 35 – 7x
(c) 40 – 7x (d) 45 – 5x

[NDA 2022 (II)]

115. How many received medals in at least two of three sports?
(a) 30–6x (b) 35–6x
(c) 40–5x (d) 40–6x

[NDA 2022 (II)]

116. How many received medals in exactly one of three sports?
(a) 21x–40 (b) 21x–35
(c) 20x–35 (d) 20x–25

[NDA 2022 (II)]

117. Consider the following statements:
1.The set of all irrational numbers between $\sqrt{2}$ *and* $\sqrt{5}$ is an infinite set.
2.The set of all odd integers less than 100 is a finite set.
Which of the following statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA 2022 (II)]

118. Let R be a relation from N to N defined by R = {(x,y) : x, y ∈ N and $x^2 = y^3$}. Which of the following are not correct?
1.(x,x) ∈ R for all x∈N
2.(x,y) ∈ R ⇒(y, x)∈R
3.(x,y)∈R and (y, z) ∈R ⇒(x,z)∈R
Select the correct answer using the code given below:
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA 2022 (II)]

119. Consider the following:
1.$A \cap B = A \cap C \Rightarrow B = C$
2.$A \cup B = A \cup C \Rightarrow B = C$
Which of the above is/are correct?
(a)1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

[NDA 2022 (II)]

120. The Cartesian product A × A has 16 elements among which are (0,2) and (1, 3). Which of the following statements is/are correct?
1.It is possible to determine set A
2.A × A contains the element (3, 2)
Select the correct answer using the code given below:
(a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

[NDA-2023 (2)]

121. Let A = [1,2,3, …..,20]. Define a relation R from A to A by R = [(x, y) : 4x – 3y = 1], where x, y ∈ A. Which of the following statements is/are correct?
1.The domain of R is {1,4,7,10,13,16}
2.The range of R is {1,5,9,13,17}
3.The range of R is equal to codomain or R
Select the correct answer using the code given below:
(a) 1 only (b) 2 only
(c) 1 and 2 (d) 2 and 3

[NDA-2023 (2)]

122. Consider the following statements:
1. A = (A ∪ B) ∪ (A – B)
2. A ∪ (B – A) = (A ∪ B)
3. B = (A ∪ B) – (A – B)
Which of the statements given above are correct?
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

[NDA-2023 (2)]

123. A set S contain (2n+1) elements. There are 4096 subsets of S which contain at most n elements. What is n equal to?
(a) 5 (b) 6
(c) 7 (d) 8

[NDA-2023 (2)]

124. Let R be a relation on the open interval (–1, 1) and is given by
$R = \{(x, y) : |x + y| < 2\}$. Then which one of the following is correct?
(a) R is reflexive by neither symmetric nor transitive
(b) R is reflexive and symmetric but not transitive
(c) R is reflexive and transitive but not symmetric
(d) R is an equivalence relation

[NDA-2024 (1)]

125. For any three non-empty sets A, B, C what is
(A ∪ B) – {(A – B) ∪ (B – A) ∪ (A ∩ B)} equal to?
(a) Null set (b) A
(c) B (d) (A ∪ B) – (A ∩ B)

[NDA-2024 (1)]

126. Let P and Q be two non-void relations on a set A. Which of the following statements are correct?
I. P and Q are reflexive ⟹ P ∩ Q is reflexive.
II. P and Q are symmetric ⟹ P ∩ Q is symmetric.
III. P and Q are transitive ⟹ P ∩ Q is transitive.
Select the answer using the code given below.
(a) I and II only (b) II and III only
(c) I and III only (d) I, II and III

[NDA-2024 (2)]

127. If A and B are two non-empty sets having 10 elements in common, then how many elements do A × B and B × A have in common ?
(a) 10 (b) 20
(c) 40 (d) 100

[NDA-2024 (2)]

128. In a class of 240 students, 180 passed in English, 130 passed in Hindi and 150 passed in Sanskrit. Further, 60 passed in only one subject, 110 passed in only two subjects and 10 passed in none of the subjects. How many passed in all three subjects ?
(a) 60 (b) 55
(c) 40 (d) 35

[NDA-2024 (2)]

129. Consider the following statements:
I. The set of all irrational numbers between √12 and √15 is an infinite set.

II. The set of all odd integers less than 1000 is finite set.
Select the correct answer using the code given below ?
(a) I only (b) II only
(c) Both I and II (d) Neither I nor II
[NDA-2024 (2)]

130. If A and B are non-empty subsets of a set, and A^c and B^c represent their complements, then which of the following is/are correct?
I. $A - B = B^c - A^c$
II. $A - B^c = A^c - B$
Select the answer using the code given below.
(a) I only (b) II only
(c) Both I and II (d) Neither I nor II
[NDA-2025 (2)]

131. In a class of 45 students, 34 like to play cricket and 26 like to play football. Further, each student likes to play at least one of the two games. How many students like to play exactly one game?
(a) 45 (b) 30
(c) 25 (d) 15
[NDA-2025 (2)]

132. Set X contains 3n elements and set Y contains 2n elements, and they have n elements in common. How many elements does $(X - Y) \times (Y - X)$ have?
(a) $5n^2$ (b) $4n^2$
(c) $3n^2$ (d) $2n^2$
[NDA-2025 (2)]

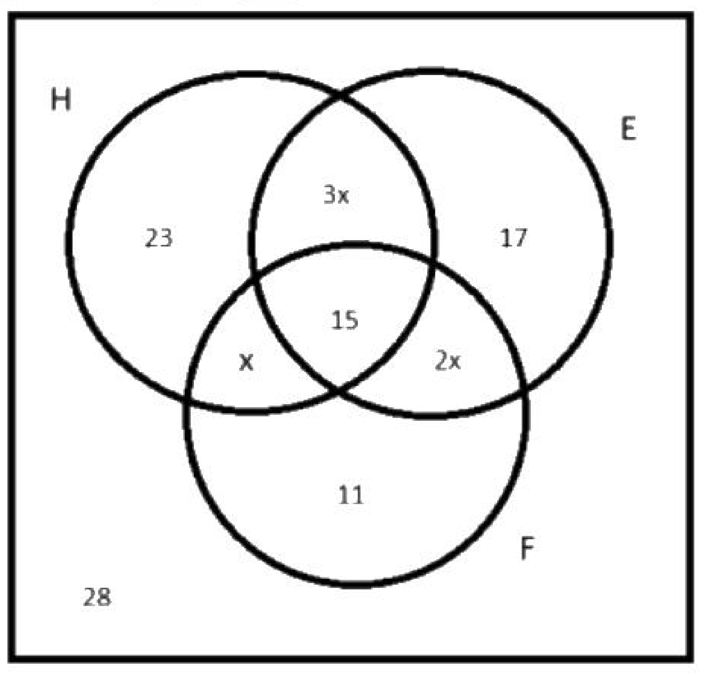
133. Let A = {–3, –2, –1, 0, 1, 2, 3} and B = {0, 1, 4, 9}. How many elements does the subset of A × B corresponding to the relation $R = \{(x, y) : |x| < y\}$ have, where $x \in A$ and $y \in B$?
(a) 9 (b) 12
(c) 15 (d) 16
[NDA-2025 (2)]

ANSWER KEY

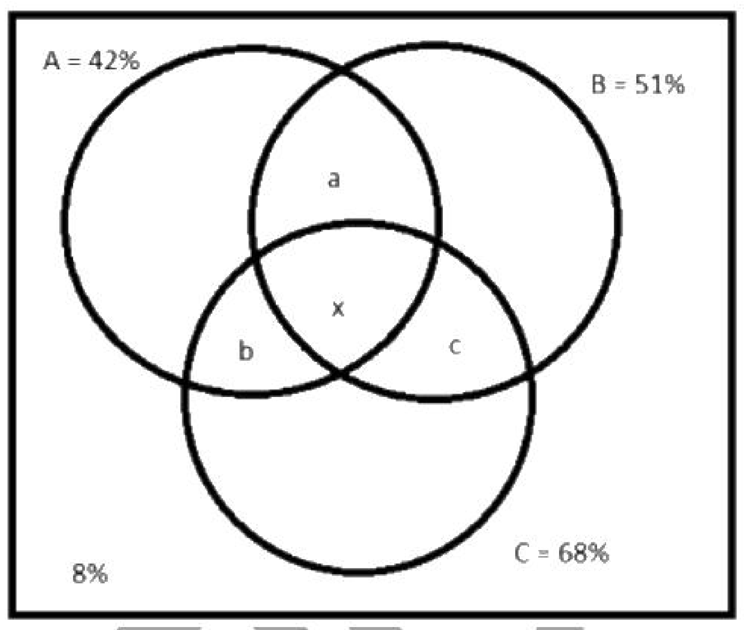
1.	b	**2.**	d	**3.**	d	**4.**	c	**5.**	d	**6.**	b	**7.**	c	**8.**	c	**9.**	c	**10.**	a
11.	a	**12.**	c	**13.**	d	**14.**	c	**15.**	d	**16.**	c	**17.**	a	**18.**	d	**19.**	b	**20.**	d
21.	b	**22**	a	**23.**	b	**24.**	c	**25.**	d	**26.**	d	**27.**	a	**28.**	b	**29.**	b	**30.**	d
31.	c	**32.**	a	**33.**	d	**34.**	d	**35.**	c	**36.**	b	**37.**	c	**38.**	c	**39.**	b	**40.**	c
41.	d	**42.**	c	**43.**	a	**44.**	c	**45.**	c	**46.**	d	**47.**	c	**48.**	a	**49.**	c	**50.**	b
51.	b	**52.**	c	**53.**	c	**54.**	d	**55.**	b	**56.**	c	**57.**	d	**58.**	b	**59.**	c	**60.**	a
61.	a	**62.**	b	**63.**	c	**64.**	b	**65.**	c	**66.**	b	**67.**	b	**68.**	a	**69.**	d	**70.**	c
71.	d	**72.**	d	**73.**	a	**74.**	b	**75.**	c	**76.**	c	**77.**	c	**78.**	b	**79.**	c	**80.**	d
81.	c	**82.**	a	**83.**	c	**84.**	c	**85.**	b	**86.**	b	**87.**	a	**88.**	d	**89.**	a	**90.**	c
91.	b	**92.**	d	**93.**	a	**94.**	b	**95.**	c	**96.**	c	**97.**	c	**98.**	c	**99.**	d	**100.**	a
101.	c	**102.**	c	**103.**	d	**104.**	b	**105.**	a	**106.**	c	**107.**	c	**108.**	c	**109.**	a	**110.**	c
111.	a	**112.**	d	**113.**	b	**114.**	c	**115.**	d	**116.**	a	**117.**	a	**118.**	d	**119.**	d	**120.**	c
121.	b	**122.**	b	**123.**	b	**124.**	d	**125.**	a	**126.**	d	**127.**	d	**128.**	a	**129.**	a	**130.**	a
131.	b																		

Solutions

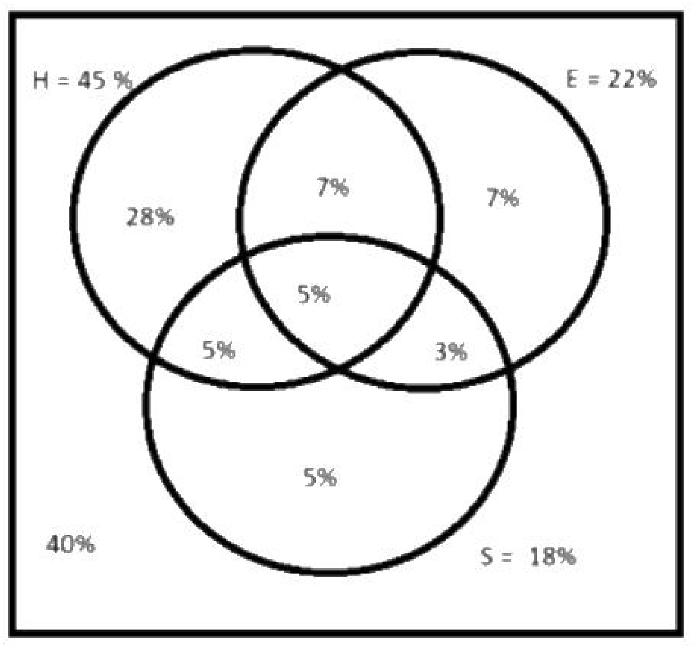
Sol.1. (b)
$\therefore$ A = {1, 2, 5, 6} and B = {1, 2, 3}
$\therefore$ A × B = {(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (5, 1), (5, 2), (5, 3), (6, 1), (6, 2), (6, 3)
And B × A = {(1, 1), (1, 2), (1, 5), (1, 6), (2, 1), (2, 2), (2, 5), (2, 6), (3, 1), (3, 2), (3, 5), (3, 6)}
$\Rightarrow$ (A × B) $\cap$ (B × A) = {(1, 1), (1, 2), (2, 1), (2, 2)}
Sol.2. (d)
$\therefore$ A, B and C are non–empty sets, such that A $\cap$ C = ϕ.
$\therefore$ (A × B) $\cap$ (C × B) = (A $\cap$ C) × B = ϕ × B = ϕ
Sol.3. (d)
$\therefore$ A = {4n + 2; n$\in$N}
= {6, 10, 14, 18, 22, 26, 30,…..}
and B = {3n : n$\in$N}
= {3, 6, 9, 12, 15, 18, 21, 24,……}
$\therefore$ A $\cap$ B = {6, 18, 30}
= {6 + (n – 1)12 | n$\in$N}
= {12n – 6 | n$\in$N}
Sol.4. (c)
Since, on the set of real numbers, R is a relation defined by xRy if and only if 3x + 4y = 5 for which
$1R\frac{1}{2}$ and $\frac{2}{3}R\frac{3}{4}$. i.e., $1R\frac{1}{2} \Rightarrow 3.1+4.\frac{1}{2}=5,$
and $\frac{2}{3}R\frac{3}{4} \Rightarrow 3\times\frac{2}{3}+\frac{3}{4}\times 4=5$
Hence, both the statements II and III are correct.
Sol.5. (d)
$\therefore$ M = Set of men and R is a relation is son of defined on M.
Reflexive relation aRa.
Since, a cannot be a son of a.
Symmetric relation
aRa $\Rightarrow$ bRa
which is also not possible.
Transitive relation aRb, aRc $\Rightarrow$ cRa
Which is not possible.
Sol.6. (b)
On PQ line and PR line , only point P is common.
Sol.7. (c)
by solving both equations $x^2 + y^2 = 4$ and x + y = 2
(0,2) and (2,0) are common points of circle and straight line.
Sol.8. (c)



In French circle
11 + x + 2x + 15 = 46
x = 5
The number of students learning precisely two languages
= x + 2x + 3x
= 6x = 6 × 5 = 30
Sol.9. (c)
by above solution
The number of students learning at least two languages
= x + 2x + 3x + 15
= 6x + 15 = 30 + 15 = 45
Sol.10. (a)
by above solution
The total strength of the class
= 28 + 23 + 17 + 11 + x + 2x + 3x + 15
= 79 + 6x + 15 = 79 + 30 + 15 = 124
Sol.11. (a)
by above solution
The number of students learn English and French
= 15 + 3x = 15 + 15 = 30
Sol.12. (c)
by above solution
The number of students learn atlest one of the languages
= 124 – 28 = 96
Sol.13. (d)
If ϕ is a null set, then its other representation is { }. i.e., ϕ = { }.
Sol.14. (c)
Given, A = {a, b, c}
Number o subsets of A = $2^n = 2^3 = 8$
{where, n = number of elements in A, n = 3}
$\therefore$ Proper subset of A = 2^n – 1 = 8 – 1 = 7
Sol.15. (d)
A $\cap$ (A $\cup$ B) = A
Sol.16. (c)
A = {1, 2, 3, 4} and B = {2, 3, 6}
A – B = {1, 4}
Complement of A – B from 1 to 10 is {2, 3, 5, 6, 7, 8, 9 ,10}
Sol.17. (a)
A = {1, 4, 9, 16, 25, 36, 49, 64, 81}
B = {2, 4, 6, 8, 10, 12, 14, 16, ……….}
A $\cap$ B = {4, 16, 36, 64} so cardinality is 4
Sol.18. (d)
n(A$\cup$B) = n(A) + n(B) + n(A$\cap$B)
330 = 260 + 210 – x
x = 140
Sol.19. (b)
If A and B have n common elements than common elements in A × B and B × A will be n^2
Sol.20. (d)
A = {0, 1} and B = {1, 0}
A × B = {(0,0), (1,0), (0,1), (1,1)} = A × A
Sol.21. (b)



n(A$\cup$B$\cup$C) = n(A) + n(B) + n(C) – n(A$\cap$B) –n(B$\cap$C) – n(A$\cap$C) + n(A$\cap$B$\cap$C)
$\Rightarrow$ n(A$\cup$B$\cup$C) = 42 + 51 + 68 – 30 – 36 – 28 + x
$\Rightarrow$ 92 = 67 + x
x = 25%
Sol.22. (a)
according to above solution
common in two sets
a + x = 30% so a = 5%
b + x = 36% so b = 11%
c + x = 28% so c = 3%
a + b + c = 19%
Sol.23. (b)
according to above solution
only A = 42 – 5 – 25 – 11 = 1
only B = 51 – 5 – 25 – 3 = 18
only C = 68 – 11 – 25 – 3 = 29
only one paper = 1 + 18 + 29 = 48%
Sol.24. (c)
according to above solution
only A = 42 – 5 – 25 – 11 = 1
Sol.25. (d)
x + 4 = 4
so x = 0
A = {0}
Sol.26. (d)
Given relation is x has the same father as y
Reflexive : x has the same father as x
Above statement is correct so relation is reflexive
Symmetric: if x has same father as y than y has same father as x, so relation is symmetric.
Transitive: If x and y have same father , y and z have same father, than x and z have same father, so relation is transitive.
Option(d) equivalence relation.
Sol.27. (a)
(1,3) is given but (3,1) is not given, so relation is not symmetric.
Sol.28. (b)
Number of elements in power set is equal to number of subsets = 2^n
Sol.29. (b)
A = {0, 1, 2} , B = {2, 3} , C = {3,4}
A $\cup$ B = {0, 1, 2, 3}
Number of elements in (A $\cup$ B) × C = 4 × 2 = 8
Sol.30. (d)
{x : x^2 + 1 = 0 } is a null set for real numbers because x^2 cannot be – 1 for real numbers.
Sol.31. (c)



by above venn diagram
People who do not know any language
= 40% of 7,50,00,000 = 3,00,00,000

Sol.32. (a)
by above venn diagram
People who know Hindi Only
= 28% of 7,50,00,000 = 2,10,00,000

Sol.33. (d)
by above venn diagram
People who know Sanskrit Only
= 5% of 7,50,00,000 = 37,50,000

Sol.34. (d)
by above venn diagram
People who know English Only
= 7% of 7,50,00,000 = 52,50,000

Sol.35. (c)
by above venn diagram
People who know only one language
= 40% of 7,50,00,000 = 3,00,00,000

Sol.36. (b)
by above venn diagram
People who know only two languages
= 15% of 7,50,00,000 = 1,12,50,000

Sol.37. (c)
B = {2, 3} and C = {3, 4}
$(B \cup C) = \{2, 3, 4|$
A = {x, y}
Number of elements in $A \times (B \cup C) = 2 \times 3 = 6$

Sol.38. (c)
a relation A on a set of R is always subset of $R \times R$

Sol.39. (b)
$A \subseteq B$, than $B^C \subseteq A^C$
For example A = {1,2} is subset of B = {1, 2,3} but complement of A is super set of B

Sol.40. (c)
If a set have 4 elements than number of elements in power set = $2^4 = 16$

Sol.41. (d)
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
So given both statements in question are wrong

Sol.42. (c)
$n(A \cup B) = n(A) + n(B) + n(A \cap B)$
50 = 30 + 40 – x
x = 20

Sol.43. (a)
Set of all even prime number is not a Null set. Because it contain one element 2.

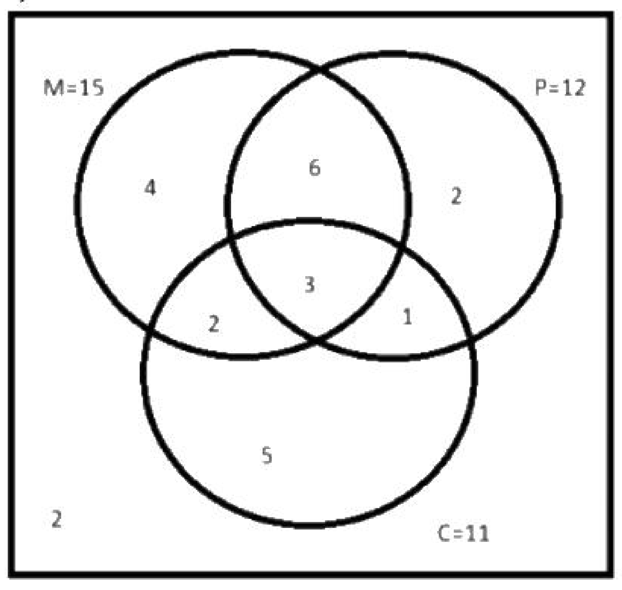
Sol.44. (c)
A = {1,2} , B = {2, 3} and C = {3, 4}
$A \times B$ = {(1, 2), (1, 3), (2, 2), (2,3)}
$A \times C$ = {(1, 3), (1, 4), (2, 3), (2,4)}
2 elements are common in $A \times B$ and $A \times C$

Sol.45. (c)
If A have n elements than number of relations from A to A = $2^{n \times n}$

Sol.46. (d)
R={(a, b): a–b is divisible by 5}
Reflexive: a – a = 0 is divisible by 5, so relation is reflexive.
Symmetric: If a – b is divisible by 5 than b – a is also divisible by 5, relation is symmetric.
Transitive: If a – b is divisible by 5 and b – c is divisible by 5, than a – c is also divisible by 5, so relation is transitive.
Option (d) relation is equivalence.

Sol.47. (c)
n(A) = 4 , n(B) = 3 than $n(A \times B) = 12$

Sol.48. (a)



By above venn diagram number of students who have taken only physics = 2

Sol.49. (c)
By above venn diagram number of students who have taken only two subjects = 2 + 6 + 1 = 9

Sol.50. (b)
The number of students who have taken only one subject =11
number of students who have taken only two subjects = 9
Statement I is wrong
The number of students who have taken at least two subjects = 12
number of students who have taken all the three subjects = 3
statement II is correct.

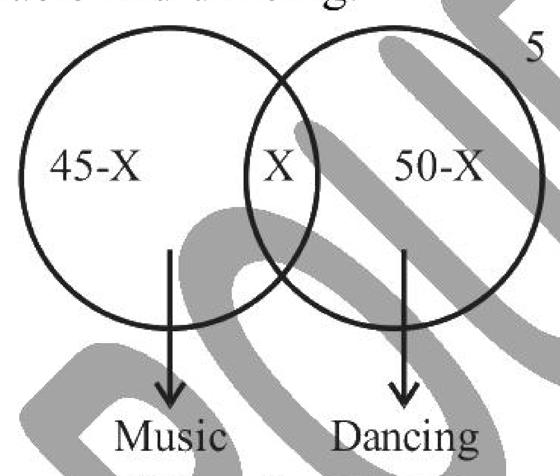
Sol.51. (b)
Relation is "difference of age of x and y is 5"
Reflexive: difference of age of x and x is zero, that is not 5
So relation is not reflexive.
Symmetric: If difference between age of x and y is 5 than difference between age of y and x is 5, relation is symmetric.
Transitive: If difference between age of x and y is 5, difference between age of y and z is 5, than difference between age of x and z will be 0 or 10, so relation is not transitive.

Sol.52. (c)
Relation is "$ab \ge 0$"
Reflexive: $a^2 \ge 0$, so relation is reflexive
Symmetric: If $ab \ge 0$ than $ba \ge 0$, relation is symmetric.
Transitive: If $ab \ge 0$ and $bc \ge 0$ than $ac \ge 0$, relation is transitive.

Sol.53. (c)
Relation is "integer x divides integer y"
Reflexive: integer x divides integer x, so relation is reflexive.
Symmetric: If integer x divides integer y than integer y does not divides integer x, relation is not symmetric.
Transitive: integer x divides integer y, integer y divides integer z than integer x always divides integer z, so relation is transitive.

Sol.54. (d)
n(A) = 5, n(B) = 4 than $n(A \times B) = 20$

Sol.55. (b)
Let the number of students in the class be x, who like both music and dancing.



$\therefore (45 - x) + x + (50 - x) + 5 = 60$
$\Rightarrow 100 - x = 60$
$\Rightarrow x = 100 - 60 = 40$

Sol.56. (c)
Required number of subsets of A containing exactly two elements
$= {}^{10}C_2 = \frac{10 \times 9}{2}$
$= \frac{90}{2} = 45$

Sol.57. (d)
A = {3, 6, 9, 12, 15, ………..}
B = {4, 8, 12, 16, 20, ………..}
C = {12, 24, 36, 48, 60, ………}
$A \cap B$ = {12, 24, 36, 48, 60, ………} = C
So $(A \cap B) - C$ = Null set = $(A \cap B) \setminus C$

Sol.58. (b)
Relation is "y is at least 5 years older than x"
Reflexive: x cannot be 5 years older than x, so relation is not reflexive.
Symmetric: If y is at least 5 years older than x than x is at least 5 years older than y, relation is not symmetric.
Transitive: y is at least 5 years older than x , z is at least 5 years older than y, than z is at least 5 years older than x, this statement is correct. so relation is transitive.

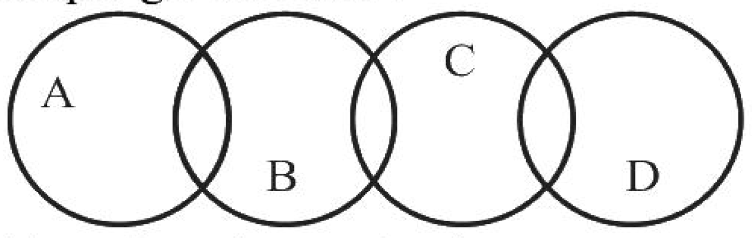
Sol.59. (c)
R={(a, b): a–b is divisible by 5}
Reflexive: a – a = 0 is divisible by 5, so relation is reflexive.
Symmetric: If a – b is divisible by 5 than b – a is also divisible by 5, relation is symmetric.
Transitive: If a – b is divisible by 5 and b – c is divisible by 5, than a – c is also divisible by 5, so relation is transitive.
Relation is equivalence.
Both statements are correct.

Sol.60. (a)
n(A) = 3, n(B) = 4 than number of relations from B to A
= $2^{12} = 4096$

Sol.61. (a)
We have,
$A = \{x \in R : x^2 + 6x - 7 < 0\}$
$= \{x \in R : (x + 7)(x - 1) < 0\}$
$= \{x \in R : -7 < x < 1\}$
and $B = \{x \in R : x^2 + 9x + 14 > 0\}$
$= \{x \in R : (x + 7)(x + 2) > 0\}$
$= \{x \in R : x < -7 \text{ and } x > -2\}$

$\therefore A \cap B = \{x \in R: -2 < x < 1\}$
$= (-2, 1)$
and $(A \setminus B) = A - B$
$= \{x \in R: -7 < x \le -2\}$
$= (-7, -2)$
Hence, only Statement I is correct.

Sol.62. (b)
Since, $A \cap B = \phi$ and $C \cap D = \phi$
$\therefore (A \cap B) \cap (C \cap D) = \phi$
$\Rightarrow (A \cap C) \cap (B \cap D) = \phi$
Hence, $A \cap C$ and $B \cap D$ may be or not disjoint.
See example given below.



Here, $(A \cap C) \cap (B \cup D) = \phi$
But, $(A \cup C) \cap (B \cup D) \neq \phi$
Hence, only Statement II is correct.

Sol.63. (c)
Relation is "y is at most 5 years older than x"
Reflexive: x cannot be at most 5 years older than x, so relation is not reflexive.
Symmetric: If y is at most 5 years differ than x than x is also at most 5 years differ than y, relation is symmetric.
Transitive: If y is at most 5 years differ than x, If z is at most 5 years differ than y, than z is at most differ than x, this statement is always not correct. For example x = 6 years , y is 9 years and z is 5 years.
so relation is transitive.

Sol.64. (b)
Relation is " $x \le y^2$ " for x and y positive numbers.
Reflexive: " $x \le x^2$ " but this expression is not correct for $0 < x < 1$. So relation is not reflexive.
Symmetric: If $x \le y^2$ than we cannot say that $y \le x^2$, for example take x = 1 and y = 2, relation is not symmetric.
Transitive: If $x \le y^2$ and $y \le z^2$, than we can say $x \le z^2$, so relation is transitive for positive numbers.

Sol.65. (c)
Relation is " n is a factor of m".
Reflexive: n is a factor of n , each natural number is a factor of itself. So relation is reflexive.
Symmetric: If n is a factor of m than we can not say that m is a factor of m. for example take n = 2 and m = 4. Relation is not symmetric.
Transitive: If n is a factor of m and m is a factor of k, than n is always a factor of k. for example take n = 2, m = 4, k = 8. Relation is transitive.

Sol.66. (b)
Divisible by 10 = n(A) = 100
Divisible by 15 = n(B) = 66
Divisible by 25 = n(C) = 40
Divisible by both 10 and 15 = Divisible by 30 = $n(A \cap B) = 33$
Divisible by both 15 and 25 = Divisible by 75 = $n(B \cap C) = 13$
Divisible by both 10 and 25 = Divisible by 50 = $n(A \cap C) = 20$
Divisible by 10, 15 and 25 = Divisible by 150 = $n(A \cap B \cap C) = 6$
Divisible by at least one out of 10, 15, 25 is $n(A \cup B \cup C)$
$n(A\cup B\cup C) = n(A) + n(B) + n(C) - n(A\cap B) - n(B\cap C) - n(A\cap C) + n(A\cap B\cap C)$
$n(A\cup B\cup C) = 100 + 66 + 40 - 33 - 13 - 20 + 6 = 146$
neither divisible by 10 nor 15 nor 25 = 1000 – 146 = 854.

Sol.67. (b)
We have,
S = set of all distinct numbers of the form p/q
where $p, q \in \{1, 2, 3, 4, 5, 6\}$
when, $p = 1, q \in \{1, 2, 3, 4, 5, 6\}$
$p = 2, q \in \{1, 3, 5\}$
$p = 3, q \in \{1, 2, 4, 5\}$
$p = 4, q \in \{1, 3, 5\}$
$p = 5, p \in \{1, 2, 3, 4, 6\}$
$p = 6, q \in \{1, 5\}$
$\therefore$ Total number of elements
= 6 + 3 + 4 + 3 + 5 + 2 = 23

Sol.68. (a)
Given, $A = \{x \in R : x^2 + 6x - 7 < 0\}$
and $B = \{x \in R: x^2 + 9x + 14 > 0\}$
$\therefore A = \{x \in R: (x + 7)(x - 1) < 0\}$
$\Rightarrow A = -7 < x < 1$
and $B = \{x \in R: (x + 2)(x + 7) > 0\}$
$B = x < -7$ or $x > -2$
$\therefore A \cap B = -2 < x < 1 = \{x\in R: -2 < x < 1\}$
$A \setminus B = -7 < x \le -2$
$= \{x \in R: -7 < x \le -2\}$
Hence, only I is correct

Sol.69. (d)
A = {HHH}
B = {HTT, THT, TTH}
C = {HHT, HTH, THH, HHH}
Option (d)
$A \cap (B' \cup C') = A \cap (B \cap C)' = A \cap \{\phi\}' = A \cap U = A$
And $B' \cap C' = (B \cup C)' = A$

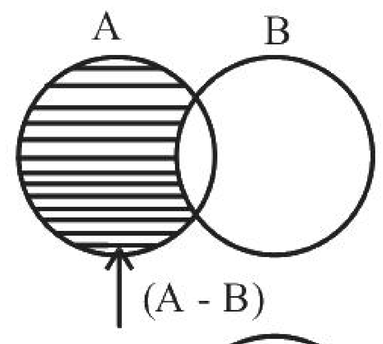
Sol.70. (c)
R = {(1,3), (1,5), (2,3), (2,5), (3,5), (4,5)}
R^{-1} = {(3,1), (5,1), (3,2), (5,2), (5,3), (5,4)}
$RoR^{-1} = R(R^{-1}) = R$ [{(3,1), (5,1), (3,2), (5,2), (5,3), (5,4)}]
= {(3,3), (3,5), (5,3), (5,5)}

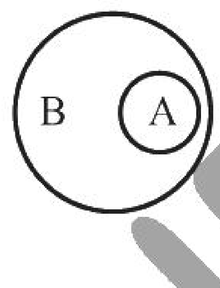
Sol.71. (d)
Maximum students can fail in all subjects is 15% and minimum is 0%

Sol.72. (d)
$S = \{x : x^2 + 1 = 0, x \text{ is real}\}$
$x^2 + 1 = 0 \Rightarrow x^2 = -1 \Rightarrow x = \sqrt{-1}$, which is not real.
$\therefore S = \phi$ or any empty set.

Sol.73. (a)
S = {All persons living in Delhi}
Let $x, y, z \in S$ such that x R y : x and y are born on the same day.
Clearly, x R x is a reflexive relation.
$x R y \Leftrightarrow y R x$, so it is symmetric relation.
If x R y and y R z $\Leftrightarrow$ x R z satisfies, so it is transitive relation.
$\therefore$ The relation is an equivalent relation.

Sol.74. (b)





1. $(A - B) \cup B = A \cup B$
So, 1 is incorrect
2. $(A - B) \cup A = A$
So, 2 is correct.
3. $(A - B) \cap B = \phi$
So, 3 is correct.
4. $A \subseteq B \Rightarrow A \cup B = B$
So, 4 is correct.

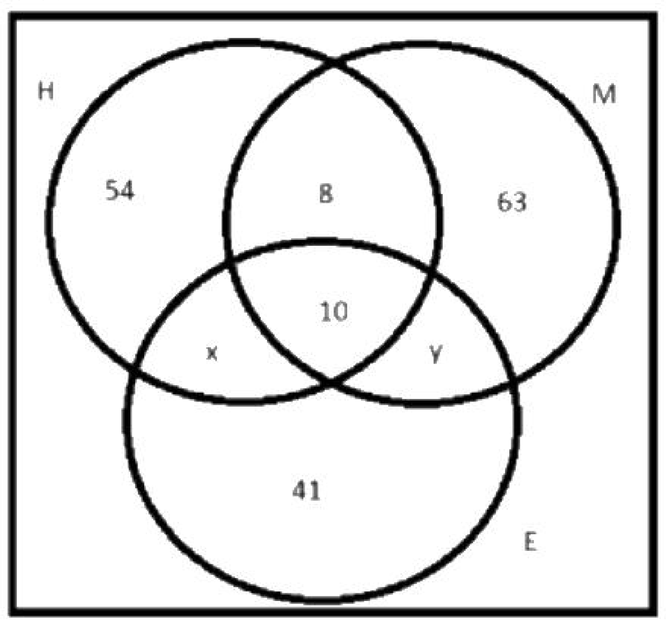
Sol.75. (c)
$\Rightarrow E - (E - (E - (E - A)))$
$= E - (E - (E - (E - A')))$
$= E - (E - (E - A))$
$= E - (E - A') = E - A$
$= A' = (B \cup C)' = B' \cap C'$

Sol.76. (c)
A = {x : x is a multiple of 2}
= {2, 4, 6, 8, …….}
B = {x : x is a multiple of 5}
= {5, 10, 15, 20, …….}
C = {x : x is a multiple of 10}
= {10, 20, 30, 40, …..}
$\therefore A \cap (B \cap C)$ = {x : x is multiple of 2, 5 and 10}
= {10, 20, 30, 40, ……} = C

Sol.77. (c)
Relation is (a,b) R (c,d) = a + d = b + c
Reflexive: (a,b) R (a,b) = a + b = b + a
Relation is reflexive.
Symmetric: (a,b) R (c,d) = a + d = b + c
Than (c,d) R (a,b) = c + b = d + a
Relation is symmetric.
Transitive: (a,b) R (c,d) = a + d = b + c ……(i)
(c,d) R (e,f) = c + f = d + e ………..(ii)
And we have to prove
(a,b) R (e,f) = a + f = b + e
And we can get this equation by adding (i) and (ii). Relation is transitive.
So relation is equivalence.

Sol.78. (b)
$n(A\cup B) = n(A) + n(B) - n(A\cap B)$
850 = 680 + 215 – x
x = 45

Sol.79. (c)



Good in either Hindi or mathematics = 54 + 8 + 63 = 125

Sol.80. (d)
By above Venn diagram.
Good in Hindi and mathematics but not in English = 8

Sol.81. (c)

$(A' \cap B) \cup (A \cap B') = A \Delta B = (A \cup B) - (A \cap B)$

Sol.82. (a)

Let x students like to play all three games

$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

$300 = 125 + 145 + 90 - [n(A \cap B) - n(A \cap B \cap C) + n(B \cap C) - n(A \cap B \cap C) + (A \cap C) - n(A \cap B \cap C)] + n(A \cap B \cap C)$.

$\Rightarrow 300 - (360 - 32 + 2x)$

$\Rightarrow 2x = 28$

$\Rightarrow x = 14$

Sol.83. (c)

Exactly one game = 300 – 32 – 14 = 254

Sol.84. (c)

$\Rightarrow A' \cup (B \cup C) \neq (C' \cap B)' \cap A'$

Sol.85. (b)

$\Rightarrow n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

$\Rightarrow n(A \cup B \cup C) = 60 + 50 + 48 - 12 - 15 - 20 + n(A \cap B \cap C)$

$\Rightarrow n(A \cup B \cup C) = 111 + n(A \cap B \cap C)$

minimum number of students = 111

Sol.86. (b)

maximum number of students = 123

because maximum value of $n(A \cap B \cap C)$ may be 12.

Sol.87. (a)

Relation is reflexive because {(1,1), (2,2), (3,3)}

Relation is symmetric because {(1,2), (2,1), (2,3), (3,2)}

Relation is not transitive because (1,3) is not there.

Sol.88. (d)

for reflexive $xRx = x^2 - 4xx + 3x^2 = 0$

$xRx = 0 = 0$ relation is reflexive

for symmetric $xRy = x^2 - 4xy + 3y^2 = 0$

$= (x - y)(x - 3y) = 0$

$= x = y$ or $x = 3y$ given

$yRx = y^2 - 4xy + 3x^2 = (x - y)(y - 3x)$

it will be zero if x = y but this result is not confirmed, relation is not symmetric with same reason relation is not transitive.

Sol.89. (a)

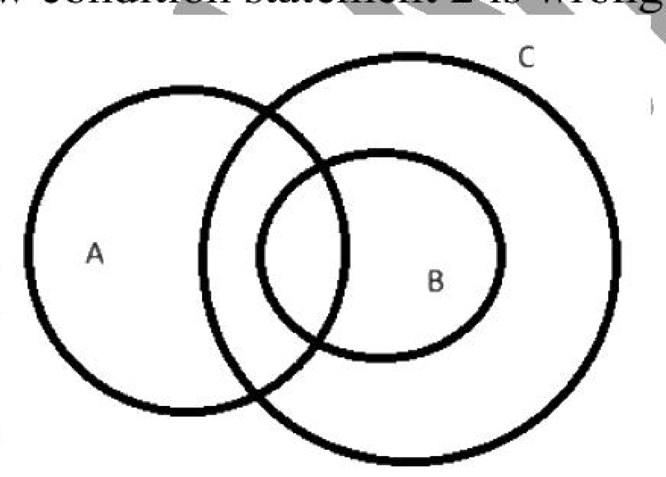
1. $(A \cap B) \cup (A \cap B') \cup (A' \cap B) = A \cup B$
2. $(A \cup (A' \cap B')) = (A \cup A') \cap (A \cup B')$

$= U \cap (A \cup B') = (A \cup B')$

Only statement 1 is correct.

Sol.90. (c)

For below condition statement 2 is wrong



Sol.91. (b)

If $A = \{\lambda, \{\lambda, \mu\}\}$, then the power set of $A = \{\phi, \{\lambda\}, \{\{\lambda, \mu\}\}, \{\lambda, \{\lambda, \mu\}\}\}$

Sol.92. (d)

number of subset = 2^n

in set S there is 10 elements

number of subset = $2^{10} = 1024$

Sol.93. (a)

$A \cup (A \cap B) = A$

Option (a) is not correct.

Sol.94. (b)

If a set A have 3 elements and set B have 6 elements than minimum number of elements in $A \cup B$ will be 6 when 3 elements are common in A and B.

Sol.95. (c)

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

= 50 + 40 – 10 = 80% play at least one game

So 20% play neither cricket nor football.

Sol.96. (c)

A = {0, 1, 2}

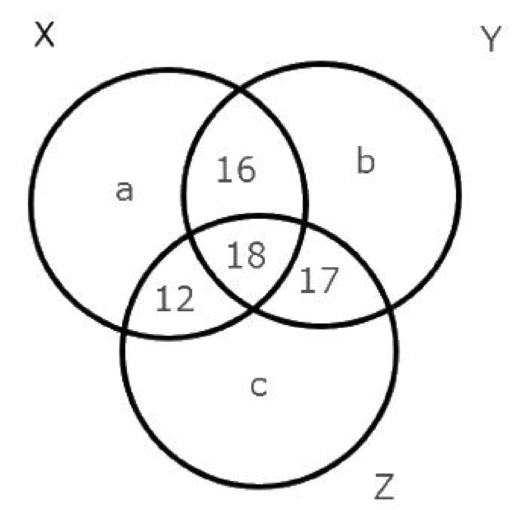
B = {2, 3, 5, 7, 11, 13, 17, 19, ……}

$A \cap B = \{2\}$

Sol.97. (c)

There is 15 proper subsets of {1, 2, 3, 4} out of these 15 superset of {3} are {3}, {1,3}, {2,3}, {4,3}, {2,3,4}, {1,3,4}, {1,2,3} that is 7.

Sol.98. (c)



Y : Z = 4 : 5

If Z = 90 than Y = 72

16 + 17 + 18 + b = 72

Than b = 21

And 12 + 18 + 17 + c = 90, than c = 43

Sol.99. (d)

$n(X) + n(Y) + n(Z) - n(X \cap Y) - n(Y \cap Z) - n(Z \cap X) + n(X \cap Y \cap Z) = n(X \cup Y \cup Z)$ = elements in at least one circle in above Venn diagram

= a + 16 + 18 + 12 + b + 17 + c

= a + b + c + 63 = a + b + 43 + 63 = a + b + 106

Sol.100.(a)

By above Venn diagram

Complement of X = b + 17 + c + p

= b + 17 + 43 + p = p + b + 60

Sol.101.(c)

$Log_a x > Log_a x$,

Above expression is wrong so relation is not reflexive.

If $Log_a x > Log_a y$ than $Log_a y > Log_a x$ is not possible so relation is not symmetric.

If $Log_a x > Log_a y$ and $Log_a y > Log_a z$ than $Log_a x > Log_a z$, so relation is transitive.

Sol.102.(c)

n(A) = n(B) in statement 1 so A and B are equivalent sets.

A = B in statement 2 so A and B are equal sets.

Both statements are correct.

Sol.103.(d)

1. The null set is a subset of every set.
2. Every set is a subset of itself.
3. If a set have 10 elements than number of elements in power set of A = $2^{10} = 1024$.

All three statements are correct.

Sol.104.(b)

2x + 3y = 20

Three Pairs of (x, y) satisfying above equation are (1,6), (4,4), (7,2) where x, y are natural numbers

Sol.105.(a)

1. statement 1 is distributive property.
2. statement 2 is De Morgan law.
3. statement 3 is wrong because if A is subset of B than A – B can also be Null set, its not necessary that A = B

So only statement 1 and 2 is correct.

Sol.106.(c)

$log_a x > log_a y$, when a = 1/2

then a < b

and this relation is not reflexive , not symmetric.

n(y) = 5

n(z) = 4

n(x–y) = 6

$n(s) = n((x-y) \cup z)$

n (s) = 10

number of proper subsets=$2^{10} - 1 = 1023$

Sol.107.(c)

A = {3, 6, 9, 12, 15, 18, ……..747, 750}

B = {2, 4, 6, 8, ………..398, 400}

$A \cap B$ = common numbers of A and B divisible by 6

$n(A \cap B) = 66$

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

= 250 + 200 – 66 = 384

Sol.108.(c)

There are 16 subsets of {1, 2, 3, 4} out of these 16 subsets, superset of {4} are {4}, {1,4}, {2,4}, {3,4}, {1,2,4}, {1,3,4}, {2,3,4} and {1,2,3,4}.

Sol.109.(d)

1. $x \notin (A \cup B) \Rightarrow x \notin A$ or $x \notin B$

Above statement is wrong

because if $x \notin (A \cup B) \Rightarrow x \notin A$ and $x \notin B$

2. $x \notin (A \cap B) \Rightarrow x \notin A$ and $x \notin B$

Above statement is wrong

because if $x \notin (A \cap B)$ than x may belongs to any one set A or B

Sol.110.(c)

1. $A \cup B = A \cap B$ if A = B
2. $A \Delta B = \varphi$ if A = B

Both above statements are correct.

Sol.111.(a)

for reflexive $xRx = x^2 - 5xx + 4x^2 = 0$

$xRx = 0 = 0$ relation is reflexive

for symmetric $xRy = x^2 - 5xy + 4y^2 = 0$

$= (x - y)(x - 4y) = 0$

$= x = y$ or $x = 4y$ given

$yRx = y^2 - 5xy + 4x^2 = (x - y)(y - 4x)$

it will be zero if x = y but this result is not confirmed relation is not symmetric with same reason relation is not transitive.

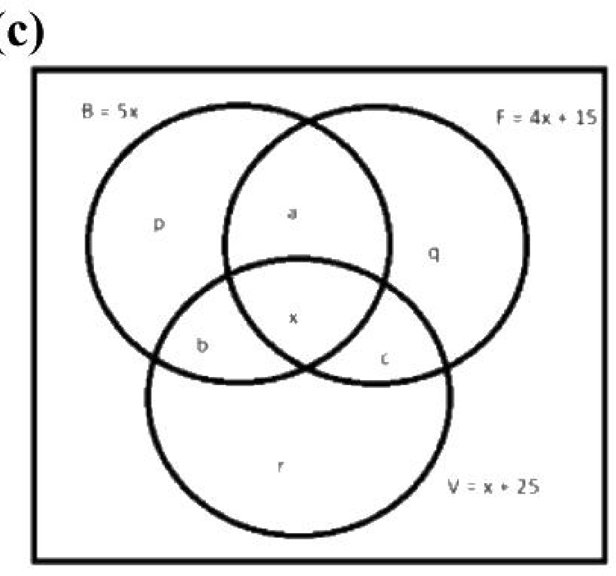
Sol.112.(d)

If R is reflexive, symmetric and transitive than R^{-1} is also reflexive, symmetric and transitive respectively.

Sol.113.(b)

There is only one element in A so number of elements in power set of A = $2^1 = 2$

Sol.114.(c)



$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

15x = 5x + (4x + 15) + (x + 25) – (a + x) – (b + x) – (c + x) + x

a + b + c = 40 – 7x

Sol.115.(d)
medals received in at least two games
$= a + b + c + x = 40 - 6x$

Sol.116.(a)
Medal received in exactly one $= 15x - (40 - 6x)$
$= 21x - 40$

Sol.117.(a)
1. There are always infinite rational and irrational numbers between any two real numbers.
2. Odd integers less than 100 are 99, 97, 95, 93, …………5, 3, 1, –1, –3, –5, ………. , this is also an infinite set.

Only statement 1 is correct.

Sol.118.(d)
Relation between and y is $x^2 = y^3$
Reflexive: $x^2 = x^3$ is only possible for x = 1. So relation is not reflexive.
Symmetric: If $x^2 = y^3$ than we cannot say that $y^2 = x^3$. So relation is not symmetric.
Transitive: If $x^2 = y^3$ and $y^2 = z^3$ than $x^2 = z^{9/2}$. So relation is not transitive.
All three statements are incorrect.

Sol.119.(d)
1. $A \cap B = A \cap C \Rightarrow B = C$
above statement 1 is incorrect.
Take A = {1,2}, B = {2,3} and C = {2,4}
2.$A \cup B = A \cup C \Rightarrow B = C$
above statement 2 is incorrect.
Take A = {1,2,3}, B = {2,3} and C = {2,4}
Both statements are wrong.

Sol.120.(c)
If A × A have 16 elements , i.e. A have 4 elements {1, 2, 3, 4}.
Both statements are correct.

Sol.121.(b)
R = {(1,1), (4,5), (7,9), (10,13), (13,17)}
Domain = {1,4,7,10,13}
Range = {1,5,9,13,17}
Codomain = {1, 2, 3, …….20}
Only 2nd statement is correct.

Sol.122.(b)
$(A \cup B) \cup (A - B) = (A \cup B)$
Statement 1 is incorrect.

Sol.123.(b)
Number of elements in set S = 2n + 1
Total subsets of S are 2^{2n+1}
Subsets containing 0 elements are ${}^{2n+1}C_0$
Subsets containing 1 elements are ${}^{2n+1}C_1$
Subsets containing 2 elements are ${}^{2n+1}C_2$
……………………..
Subsets containing n elements are ${}^{2n+1}C_n$
Given that
${}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + \ldots\ldots + {}^{2n+1}C_n = 4096$
[sum of complete combination series is 2^n and sum of half combination series is 2^{n-1}, but in this question number of elements are 2n + 1]
$2^{2n+1-1} = 4096 = 2^{12}$
2n = 12
n = 6

Sol.124.(d)
Relation is $|x + y| < 2$ for $-1 < x, y < 1$
Reflexive: $|x + x| < 2$
$|2x| < 2$
$|x| < 1$, relation is reflexive.
Symmetric: $|x + y| < 2$, than $|y + x| < 2$. relation is symmetric.
Transitive: $|x + y| < 2$, $|y + z| < 2$ than we cannot say that $|x + z| < 2$ but for $-1 < x, y < 1$, $|x + z| < 2$ is correct. Relation is transitive.
Answer is equivalence relation.

Sol.125.(a)
$(A \cup B) - \{(A - B) \cup (B - A) \cup (A \cap B)\}$
$(A \cup B) - (A \cup B)$ = Null set.

Sol.126.(d)
I. P and Q are reflexive $\Rightarrow P \cap Q$ is reflexive.
II. P and Q are symmetric $\Rightarrow P \cap Q$ is symmetric.
III. P and Q are transitive $\Rightarrow P \cap Q$ is transitive.
All above statements are correct.

Sol.127.(d)
If A and B are two non-empty sets having 10 elements in common, then number of common elements in A × B and B × A $= 10^2 = 100$

Sol.128.(a)
$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$
$230 = 180 + 130 + 150 - [n(A \cap B) - n(A \cap B \cap C) + n(B \cap C) - n(A \cap B \cap C) + (A \cap C) - n(A \cap B \cap C)] + n(A \cap B \cap C)$.
$230 = 180 + 130 + 150 - 110 - 2n(A \cap B \cap C)$.
$2n(A \cap B \cap C) = 120$
$n(A \cap B \cap C) = 60$

Sol.129.(a)
1. There are always infinite rational and irrational numbers between any two real numbers.
2. Odd integers less than 1000 are 999, 997, 995, 993, …………5, 3, 1, –1, –3, –5, ………. , this is also an infinite set.

Sol.130.(a)
I. $B^c - A^c = (U - B) - (U - A) = A - B$
II. $A - B^c = A - (U - B) = B - A^c$
Statement I is correct, II is incorrect

Sol.131.(b)
$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
let x students like to play both games
45 = 34 + 26 – x
x = 15
Students who play only cricket = 34 – 15 = 19
Students who play only football = 26 – 15 = 11
Students like to play exactly one game = 30

Sol.132.(d)
No of elements in $(X - Y) = 3n - n = 2n$
No of elements in $(Y - X) = 2n - n = n$
No of elements in $(X - Y) \times (Y - X)$
$= 2n \times n = 2n^2$

Sol.133.(a)